KeloidBench: Interpretable machine learning on fibroblast program features for keloid detection

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Abstract

We present KeloidBench, a demonstration pipeline for predicting whether a tissue transcriptomic profile is *keloid* versus *unaffected/normal skin* under true cross-study generalization. KeloidBench collapses high-dimensional gene expression into compact fibroblast activation program features motivated by a replication-first observation: single marker genes are unstable across public cohorts [Naitoh et al., 2005, Fuentes-Duculan et al., 2017], while aggregated fibroblast-state programs transfer more reliably [Deng et al., 2021]. Moreover, KeloidBench acts as an auditable GenAI copilot for cross-cohort keloid transcriptomic analysis. The model initializes a deterministic program-score to produces predictions, confidence estimates, abstention decisions, and feature contributions, and follows up with a constrained language framework that converts these structured outputs into traceable biological summaries without generating or modifying the underlying predictions. We evaluate linear classifiers with leakage-safe nested leave-one-study-out (LOSO) selection, accession$\times$donor-balanced weighting, optional Platt-calibrated probabilities [Platt, 1999], and donor-grain scoring when donor labels are class-pure. On a transferable subset of public studies, program features achieve approximately 0.92 LOSO weighted F1, with approximately 0.84 pooled accuracy outside the LOSO setting. These results support KeloidBench as a research-facing public-cohort triage tool and motivate prospective matched-cohort or biobank validation before any clinical use.

1 Introduction

Keloids are pathologic scars whose molecular state is often studied with transcriptomics. They are a major concern in plastic surgery and cosmetic dermatology because they can cause disfigurement, symptoms (e.g., pruritus or pain), and high recurrence after excision, directly affecting aesthetic and functional outcomes [Rajadurai and Tsao, 2024, Latoni and Tsao, 2024]. They also matter in military and trauma medicine, where burns, blast injuries, and complex wounds make scar modulation and durable healing central to long-term recovery and readiness.

A recurring challenge is that reported marker genes and hub-gene signatures do not reliably replicate across public cohorts and platforms, which complicates building models that generalize [Latoni and Tsao, 2024]. Despite this importance, keloid molecular datasets are small, heterogeneous, and inconsistently labeled. Across studies, "keloid" may refer to lesional tissue, perilesional margins, or postsurgical scar collected at different anatomical sites and time points, profiled on different platforms, and processed with different normalization choices. In addition, pathology determination itself is imperfect: boundaries between hypertrophic scar, keloid, and other fibrotic or inflammatory dermal states can be subtle, and routine histopathology is variably reported or unavailable in public repositories. These factors induce label noise and domain shift that can overwhelm models tuned to a single cohort [Kang et al., 2020].

Preprint.

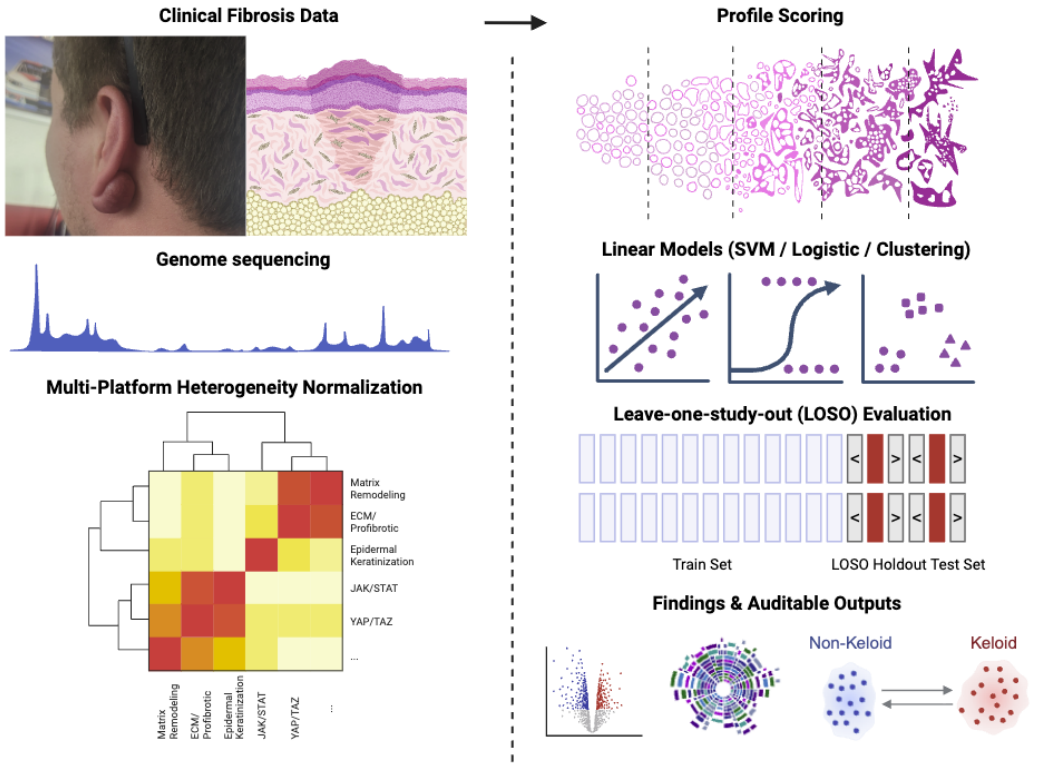


Figure 1: End-to-end KeloidBench pipeline for robust, multi-study keloid fibrosis modeling: heterogeneous public transcriptomic profiles are harmonized into stable fibroblast program scores (e.g., ECM/profibrotic, remodeling, JAK/STAT, YAP/TAZ), scored per sample, and fed to interpretable linear classifiers with calibrated probabilities and abstention; strict leave-one-study-out (LOSO) validation quantifies true cross-study generalization and yields auditable outputs (per-profile predictions, feature attributions, and cohort-level summaries) for research triage and downstream matched-cohort validation.

A dedicated benchmark is therefore needed to make progress measurable and comparable. Such a benchmark should (i) define a clear primary endpoint (keloid versus unaffected/normal skin), (ii) standardize metadata and inclusion/exclusion criteria to avoid inadvertent label mixing (e.g., perilesional versus lesional), (iii) enforce study-level holdouts to quantify true cross-study generalization, and (iv) report calibrated uncertainty and abstention to reflect ambiguous pathology rather than forcing overconfident predictions. Without these guardrails, reported "marker genes" and signatures can look strong in-sample yet fail to replicate when evaluated on new studies.

Machine learning models are well-suited to this setting because they can integrate weak, distributed transcriptomic signals into a single decision function while explicitly modeling uncertainty. By training and evaluating under study holdout, a model can be used as a reproducible, auditable baseline for cross-cohort comparison, and as a research triage tool to flag samples whose molecular state is inconsistent with their nominal label for follow-up review (e.g., pathology re-check, sample swap, anatomical mismatch). We seek to answer whether a machine learning model accurately predict whether a tissue expression profile is *keloid* versus *unaffected/normal skin* in silico under live, true study holdout conditions. This demonstration paper presents an auditable pipeline that (i) collapses high-dimensional gene expression into a small panel of fibroblast activation program scores and (ii) evaluates models under leakage-safe leave-one-study-out (LOSO) splits.

2 Related work

Keloid biology has been profiled across bulk and single-cell transcriptomic studies, revealing fibroblast heterogeneity and profibrotic signaling axes while also highlighting substantial cross-cohort variability

in reported markers [Deng et al., 2021, Fuentes-Duculan et al., 2017, Wu et al., 2020]. Despite this progress, two practical gaps limit translation to robust computational tools: (i) public cohorts are multi-platform and inconsistently curated, so reported signatures often fail to transfer without careful handling of batch/unwanted variation (e.g., ComBat and surrogate variable analysis) [Johnson et al., 2007, Leek et al., 2012], and (ii) most pipelines do not expose calibrated uncertainty and auditable rationales that help researchers decide when to trust, abstain, or investigate outliers [Platt, 1999].

Our design bridges these gaps by combining the strengths of simple, stable linear models (for cross-study generalization and transparent feature attributions) with a vision–language-model-inspired interface: structured, deterministic model outputs are surfaced in a dashboard and then translated into traceable natural-language summaries without letting the language component change the underlying predictions. This "copilot" framing aims to make multi-study transcriptomic triage both rigorous (LOSO, calibration) and usable (interpretable program-space views and narrative summaries) for downstream matched-cohort validation.

3 Methods and Experiments

System inputs/outputs. The demo takes an expression-derived feature table (e.g., a CSV/Parquet of program scores per sample) and outputs (i) per-sample predicted class probabilities for `keloid_vs_unaffected_skin`, (ii) LOSO evaluation summaries, and (iii) optional *selective triage* behavior that abstains when confidence is low.

The system is designed for researchers who want a reproducible, cross-study baseline and a concrete starting point for prospective validation (matched cohort or retrospective biobank). **Data, input representation, and task.** The raw input contains thousands of genes, but the primary model views use fixed program scores. Each program score is the mean (or within-sample rank contrast) of a small curated gene set (e.g., `POSTN`, `CTHRC1`, `COL1A1`), computed after per-accession scaling to make platforms more comparable. The primary endpoint is `keloid_vs_unaffected_skin`, and evaluation uses accession-defined studies as the unit of holdout (LOSO).

Feature views and classifier stack. KeloidBench collapses high-dimensional gene expression into compact, interpretable program features computed as means (or within-sample rank contrasts) over curated gene sets after per-accession scaling. We use linear models that are stable under LOSO with 4–16 features and on the order of $\sim$40 donors: (i) linear SVM (probabilities via Platt scaling or decision-function scaling), (ii) ridge logistic regression, and (iii) elastic-net logistic regression (L1+L2) to suppress noisy programs.

Training and evaluation architecture. Random sample splits can leak via shared accession-specific effects (platform, batch, and often donors), so we treat the study/accession as the holdout unit. To avoid large studies dominating optimization, we use accession$\times$donor-balanced sample weights and score candidate configurations by *equal-accession mean* performance in the inner loop (each inner study contributes equally). When donor identifiers are available and donors are class-pure within a study, we additionally report donor-grain metrics by aggregating predictions per donor; otherwise we fall back to sample-grain scoring.

Algorithm 1 Leakage-safe nested LOSO protocol (selection, prediction, scoring)

Require: Studies $\{S_i\}_{i=1}^{n}$; lockboxes L; views $\mathcal{V}$; models $\mathcal{M}$; grid Θ; metric m; donor map d (opt.).

for $s \in [n] \setminus L$ **do**
  $\mathcal{T} \leftarrow [n] \setminus (L \cup \{s\})$, $\mathcal{C} \leftarrow \mathcal{V} \times \mathcal{M}$.
  for $C \in \mathcal{C}$ **do**
    for $t \in \mathcal{T}$ **do**
      $C^{(-t)} \leftarrow \mathrm{fit}(C, \mathcal{T} \setminus \{t\})$; $p_{C,t}(x) \leftarrow P_{C^{(-t)}}(y{=}1 \mid x)$, $x \in S_t$.
    end for
    $p_C \leftarrow \cup_{t \in \mathcal{T}} p_{C,t}$; $\tau_C \leftarrow \arg\max_{\tau \in \Theta} m\big(y_{\mathcal{T}}, \mathbb{I}[p_C \geq \tau]\big)$.
    $\mu_C \leftarrow \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} m\big(y_t, \mathbb{I}[p_{C,t} \geq \tau_C]\big)$.
  end for
  $\{C_k, \tau_k\}_{k=1}^{K} \leftarrow \mathrm{TopK}(\mu_C)$; fit C_k on $\mathcal{T}$; $p_k(x) \leftarrow P_{C_k}(y{=}1 \mid x)$, $x \in S_s$.
  $\bar{p}(x) \leftarrow \frac{1}{K} \sum_k p_k(x)$; $\tau \leftarrow \mathrm{median}(\tau_1, \ldots, \tau_K)$; $\hat{y}(x) \leftarrow \mathbb{I}[\bar{p}(x) \geq \tau]$.
  score(S_s) at sample-grain, or donor-grain if class-pure: $\tilde{p}_u \leftarrow \mathrm{mean}_{x:d(x)=u} \bar{p}(x)$.
end for

4 Results and Demonstration

Fibroblast program features transfer across public tissue cohorts substantially better than singular marker genes. On a transferable subset of public studies, the model reaches approximately 0.92 LOSO weighted F1, and approximately 0.84 accuracy without the LOSO setup. However, due to noisy and heterogeneous public datasets, some individual data points and cohorts remain difficult to predict. Hence, these results motivate using the system as a *public-data triage* tool and to green-light a matched cohort (or retrospective biobank) study to support a clinical accuracy claim.

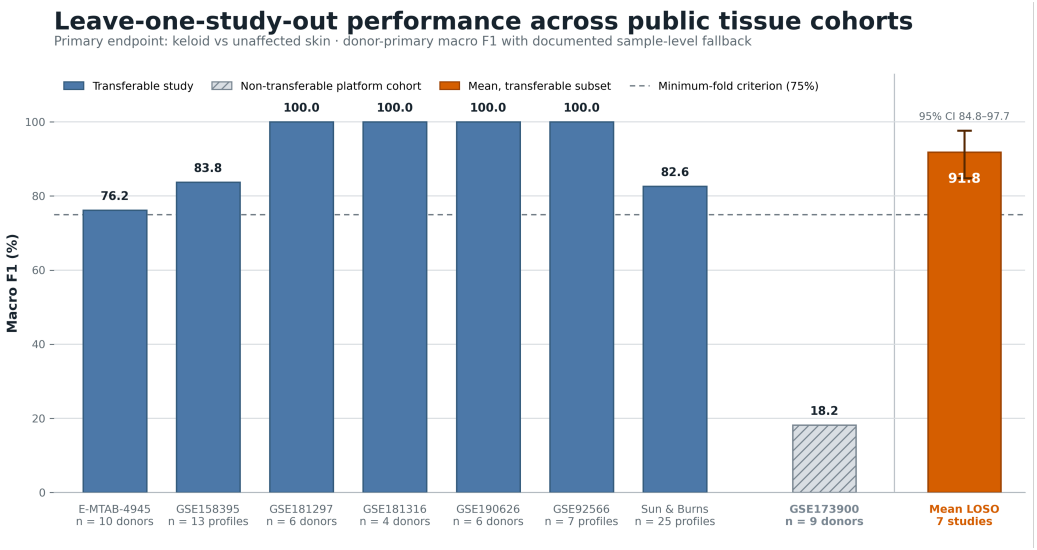


Figure 2: Cross-study performance summary under leave-one-study-out (LOSO) evaluation for `keloid_vs_unaffected_skin`, highlighting robust transfer on a subset of public cohorts and residual hard cases under public-cohort heterogeneity.

The live demo centers on an interactive web dashboard that mirrors the benchmark protocol and exposes model behavior at both the profile and study levels. We will (i) select a public transcriptomic profile and display its frozen scorer output for `keloid_vs_unaffected_skin` (predicted probability, predicted label, and calibrated confidence), (ii) vary the user-facing abstention rule (e.g., abstain when $\max(p, 1-p)$ falls below a chosen confidence threshold) to illustrate selective triage under ambiguity, and (iii) inspect interpretability views that make the linear program-feature architecture auditable: a program-activity "fingerprint" compared to class medians, a program-space map where clicking a point loads the corresponding profile, and a per-feature contribution plot showing which programs push the decision toward keloid versus unaffected skin.

In addition to browsing frozen public profiles, the dashboard supports ad hoc testing of user-provided inputs: users can upload their own expression-derived profile (or provide a GEO/GSE identifier when available) to run the same frozen scorer and receive the same set of structured outputs (probability, confidence, abstention decision, and feature contributions) for side-by-side comparison against the benchmark distribution.

To emphasize true cross-study generalization, we will then switch to study-held-out evidence: the dashboard allows browsing LOSO outer folds by held-out accession, reporting per-study weighted F1/accuracy alongside selective coverage and listing per-profile predictions (including abstentions). The demo is intentionally flexible: users can adjust abstention thresholds, swap between feature views and frozen model variants, and upload their own expression-derived profiles (or reference public accessions) to run the same scorer and compare against the benchmark distribution. For dermatologists and scar researchers, this interface provides a practical way to audit which fibroblast programs drive a decision, triage borderline cases for follow-up (e.g., pathology review or re-annotation), and adapt the workflow to related dermal fibrosis settings where labeled data are sparse. Throughout, all displayed numbers and tables are exported from versioned repository artifacts, enabling end-to-end provenance from each interactive panel back to the underlying public cohort and frozen model configuration.

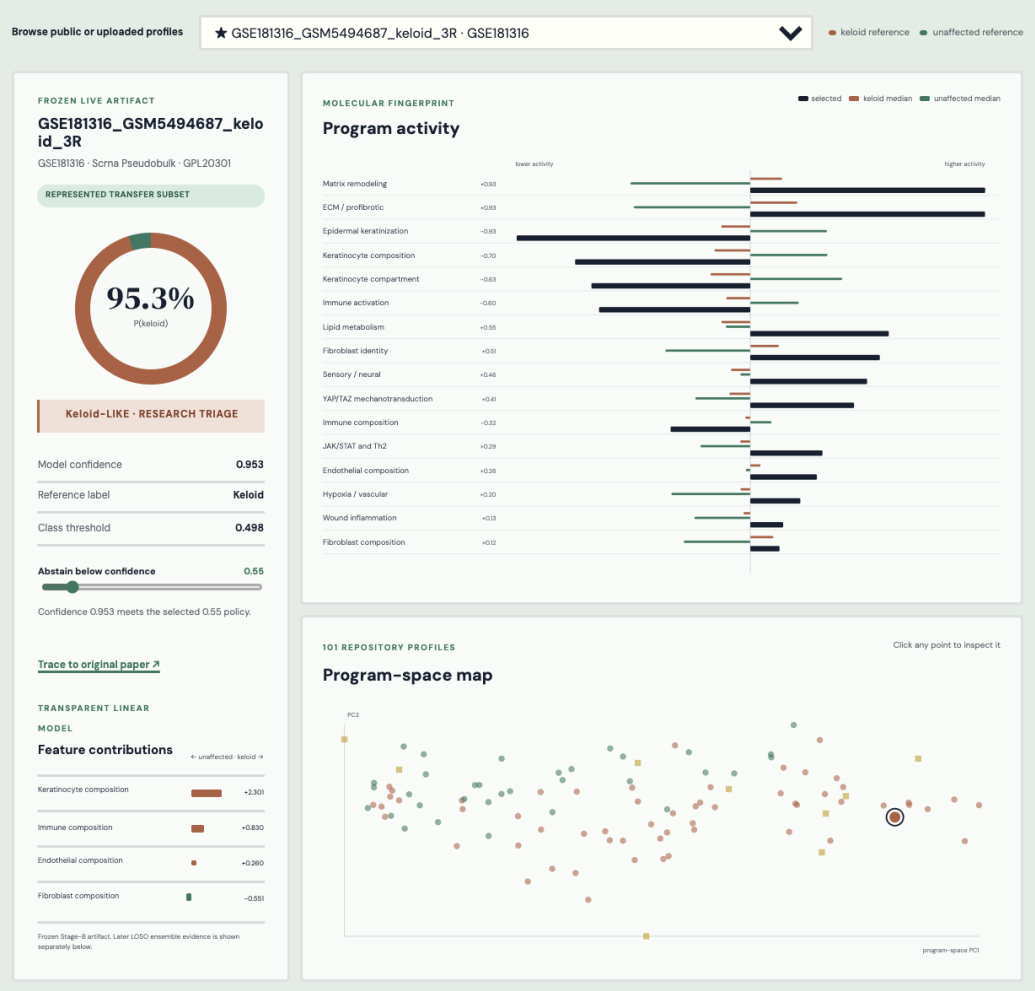


Figure 3: Live demo web dashboard (study-held-out evidence) illustrating four representative profile "stories" surfaced during LOSO browsing. The interface reports per-study weighted metrics and selective coverage under abstention, and highlights four key individual profiles as: 1) a confident keloid-like case (shown) with high live keloid score, 2) a confident unaffected-like case with high unaffected score, 3) a borderline profile that is deliberately withheld via confidence-based abstention, and 4) a known transfer failure from a non-transferable platform with an incorrect live prediction.

5 Conclusion

We presented an interpretable, program-feature classifier stack for keloid versus unaffected skin that is evaluated under strict study-held-out (LOSO) validation and surfaced through a provenance-linked dashboard. Across public transcriptomic cohorts, the model's performance and auditable feature attributions support its use as a practical *public-data triage* tool to prioritize hypotheses, refine inclusion criteria, and justify follow-on matched-cohort or retrospective biobank studies. After GEO-scale in silico evidence is established, the next step is prospective in vivo validation, ideally culminating in a controlled clinical trial to quantify clinical utility and safety in real patient workflows.

Future work should focus on improving robustness and evidence quality: expanding to larger and more diverse cohorts, strengthening batch/platform harmonization, stress-testing generalization across endpoints and study designs, and integrating additional modalities or clinical covariates where available. The system is intended for research and cohort triage rather than clinical diagnosis; public-cohort heterogeneity, platform effects, and endpoint-definition differences can yield failure modes even when average metrics are strong, so prospective validation on a matched cohort is required before any high-stakes deployment.

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